MODULE D

AI integration and chatbot projects

Three complete chatbot-family projects, from the simplest to a document assistant. Each has a test log.

Project 1: FAQ bot for a coaching centre

Four intents (greeting, price, timing, booking), a fallback handover and a test set of ten messages. The complete code is in the Chatbot developer chapter; run its tests before you publish.

PYTHON

```
import re

INTENTS = [
    ("greeting", ["hi", "hello", "hey"], "Hello! I can help with prices, timings and bookings."),
    ("price", ["price", "cost", "fee", "fees", "how much"], "Our fees start at Rs 5,000. Tell me
the course and I will confirm."),
    ("timing", ["timing", "timings", "schedule", "when"], "Classes run on weekends. Which batch do
you prefer?"),
    ("booking", ["book", "enrol", "enroll", "join", "seat"], "Great! Please share your name and
phone number and a counsellor will call you."),
]
HANDOFF = "I will connect you with a counsellor for that."

def normalise(text):
    return re.sub(r"[^a-z ]", " ", text.lower())

def reply(message):
    words = normalise(message)
    for name, keywords, answer in INTENTS:
        for kw in keywords:
            if re.search(rf"\b{re.escape(kw)}\b", words):
                return name, answer
    return "handoff", HANDOFF

if __name__ == "__main__":
    assert reply("Hi there")[0] == "greeting"
    assert reply("how much is the course?")[0] == "price"
    assert reply("Can I book a seat for Monday?")[0] == "booking"
    assert reply("When are the timings?")[0] == "timing"
    assert reply("Is there a refund policy?")[0] == "handoff"
    assert reply("this is fine")[0] == "handoff"   # 'hi' inside 'this' must not match
    print("chatbot tests passed")
```

Project 2: lead-capture bot with a CRM

1. Greeting and two questions: course and phone number.
2. Validate the phone number on the server (10 digits after removing symbols).
3. Create a lead in the CRM through its API or an automation tool.
4. Send a template confirmation within the messaging platform's rules.
5. Test with a valid lead, an invalid phone number and a duplicate.

Project 3: knowledge assistant over documents

Retrieval first, then the model answers only from the retrieved passages and cites them. The small retrieval program below shows the principle without any API key.

PYTHON

```
import re
from collections import Counter

STOP = {"the", "and", "for", "are", "you", "your", "how", "what", "does", "long", "take",
        "have", "with", "from", "that", "this", "after", "within", "each", "shown", "before",
"run", "our", "can", "will"}
DOCS = {
    "refund": "Refunds are processed within 7 working days after a written request.",
    "fees": "Course fees are shown on each program page and confirmed in writing before payment.",
    "timings": "Live classes run on Saturdays and Sundays in the evening.",
}

def tokens(text):
    return [w for w in re.findall(r"[a-z]+", text.lower()) if len(w) > 2 and w not in STOP]

def retrieve(question, docs=DOCS, k=1):
    q = Counter(tokens(question))
    scored = []
    for key, text in docs.items():
        d = Counter(tokens(text))
        score = sum(min(q[w], d[w]) for w in q)
        scored.append((score, key, text))
    scored.sort(reverse=True)
    return [(key, text) for score, key, text in scored[:k] if score > 0]

def answer(question):
    hits = retrieve(question)
    if not hits:
        return "Not found in our documents. A team member will reply."
    key, text = hits[0]
    return f"{text} [source: {key}]"

if __name__ == "__main__":
    assert answer("How long do refunds take?").endswith("[source: refund]")
    assert "7 working days" in answer("How long do refunds take?")
    assert answer("What are the fees for the course?").endswith("[source: fees]")
    assert answer("When do the live classes run?").endswith("[source: timings]")
    assert answer("Do you sell pizza?").startswith("Not found")
    print("rag tests passed")
```

Security and privacy for all three

- No personal data in logs.
- Clear notice that the visitor is talking to a bot, and how to reach a person.
- Opt-out and data deletion on request.

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